



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

Name: Abobakr Mohamed

PRN: 25070122015

Branch: CSE (2025-29)

Semester: 3

Subject: Programming with Java

ASSIGNMENT No.: 11

TITLE: Write a Java program to demonstrate the implementation of interfaces.

Problem Statement:

Write a Java program to demonstrate the implementation of interfaces.

Learning Objectives:

To implement interfaces for achieving abstraction and multiple inheritance of type.

Theory:

An interface specifies a set of abstract methods that implementing classes must define. It provides complete abstraction and supports multiple inheritance of type. Interfaces improve modularity, loose coupling, and software flexibility. They allow different classes to implement common behavior while maintaining independent implementations. Interfaces are widely used in enterprise application development.

Outcomes:

Interfaces were successfully created and implemented across multiple, unrelated classes, with each class providing its own implementation of the same abstract method(s), as verified through the sample programs.

```

interface Drawable{
    void draw();

    double PI = 3.14159;
}

class Square implements Drawable{
    double side;

    Square(double s){
        this.side = s;
    }

    public void draw(){
        System.out.println("Drawing a Sqaure with side " + side);
    }
}

class Circle implements Drawable{
    double radius;

    Circle(double r){
        this.radius = r;
    }

    public void draw(){
        System.out.println("Drawing a Circle with radius " + radius);
        System.out.println("Area using interface contstant PI: " + (PI * radius *
radius));
    }
}

public class JAVA_EXP_11{
    public static void main(String [] args){

        Drawable shape1 = new Square(5);
        shape1.draw();

        System.out.println();

        Drawable shape2 = new Circle(4);
        shape2.draw();
    }
}

```

```
in' 'JAVA_EXP_11'
```

```
Drawing a Sqaure with side 5.0
```

```
Drawing a Circle with radius 4.0
```

```
Area using interface contstant PI: 50.26544
```

```
PS C:\Users\abell\OneDrive\Desktop\JAVA_PRACTICALS>
```

Conclusion:

Interfaces provide full abstraction by defining a contract of methods without any implementation, letting unrelated classes commit to the same behaviour in their own way. They enable a form of multiple inheritance of type in Java, promote loose coupling between components, and are widely used to design flexible, modular systems where implementations can vary independently of the interface itself.

Exercise:

1. Create an interface Printable and implement it in Student and Employee classes.

```
interface Printable {
    void print();
}

class Student implements Printable {
    String name;
    int rollNo;
    double marks;

    Student(String name, int rollNo, double marks) {
        this.name = name;
        this.rollNo = rollNo;
        this.marks = marks;
    }

    @Override
    public void print() {
        System.out.println("---- Student Details----- ");
        System.out.println("Name      : " + name);
        System.out.println("Roll No : " + rollNo);
        System.out.println("Marks   : " + marks);
    }
}

class Employee implements Printable {
    String name;
    int empId;
    double salary;

    Employee(String name, int empId, double salary) {
        this.name = name;
    }
}
```

```

        this.empId = empId;
        this.salary = salary;
    }

    @Override
    public void print() {
        System.out.println("---- Employee Details----- ");
        System.out.println("Name      : " + name);
        System.out.println("Emp ID   : " + empId);
        System.out.println("Salary  : " + salary);
    }
}

public class JAVA_EXP_11 {
    public static void main(String[] args) {
        Printable p1 = new Student("Ananya Joshi", 21, 88.5);
        p1.print();

        System.out.println();

        Printable p2 = new Employee("Vikram Singh", 305, 62000.0);
        p2.print();
    }
}

```

```

C:\Users\abell\OneDrive\Desktop\JAVA_PRACTICALS> java -cp . JAVA_EXP_11
---- Student Details ----
Name      : Ananya Joshi
Roll No   : 21
Marks     : 88.5

---- Employee Details ----
Name      : Vikram Singh
Emp ID    : 305
Salary    : 62000.0
PS C:\Users\abell\OneDrive\Desktop\JAVA_PRACTICALS>

```

2. Create an interface Switchable with a method turnOn(). Implement the interface in Light and Fan classes to display the device status.

```

interface Switchable {
    void turnOn();
}

class Light implements Switchable {
    String location;
}

```

```

    Light(String location) {
        this.location = location;
    }

    @Override
    public void turnOn() {
        System.out.println("Light in " + location + " is now ON");
    }
}

class Fan implements Switchable {
    int speed;

    Fan(int speed) {
        this.speed = speed;
    }

    @Override
    public void turnOn() {
        System.out.println("Fan is now ON, running at speed level " + speed);
    }
}

public class JAVA_EXP_11 {
    public static void main(String[] args) {
        Switchable device1 = new Light("Living Room");
        device1.turnOn();

        Switchable device2 = new Fan(3);
        device2.turnOn();
    }
}

```

```

\workspaceStorage\6b4136d8ec875b8bb324a9029103fc22\redhat.java\jd
in' 'JAVA_EXP_11'
Light in Living Room is now ON
Fan is now ON, running at speed level 3
PS C:\Users\abell\OneDrive\Desktop\JAVA_PRACTICALS>

```